

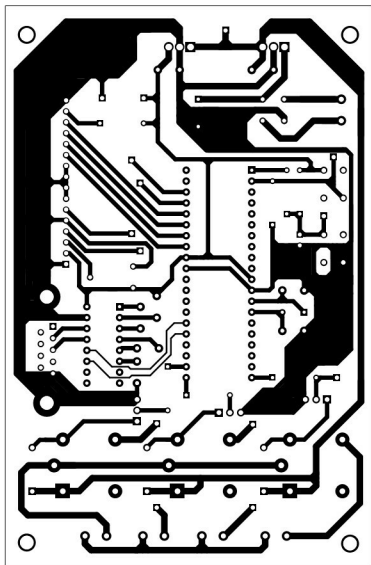


Fig. 8: An actual-size, single-side PCB layout of the circuit

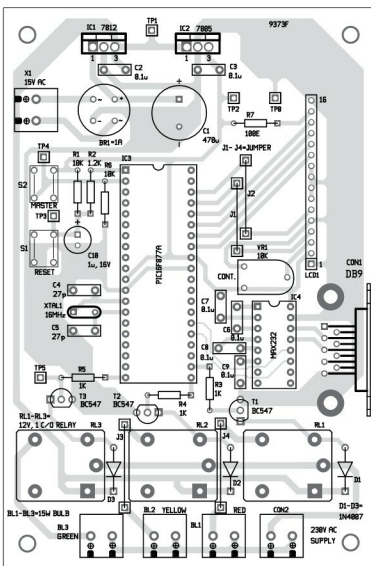


Fig. 9: Component layout for the PCB

Once the PC is connected to the microcontroller, name of the item (for example, Folk song general) going to be presented on the stage is selected from the drop-down menu in Form2. After pressing the Next button of the Form2, the item detail along with corresponding minimum, maximum and grace time fetched from MS Access file is displayed on your PC's screen. The timer will be displayed in the green box in the middle of Form3 as shown in Fig. 7.

This data is also communicated to the LCD module through microcontroller, which also displays the same timing. As soon as a team starts its presentation on the stage, Start button is pressed, the green bulb glows and running time in the green box in Form3 and LCD module keeps on changing simultaneously. When the running time is equal to minimum time of that particu-

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on source.efymag.com

lar item, the background colour of the green box timer changes from green to yellow and green bulb goes off and yellow bulb glows.

Again, when the timer value is more than the maximum time, this yellow bulb starts blinking (goes on and off for every two seconds), indicating that the grace period has started. Blinking continues till the grace period is over. After expiry of the grace period, yellow bulb goes off and red bulb glows. Then the buzzer of the system (PC/laptop) produces a continuous beep sound, unless you press the Stop

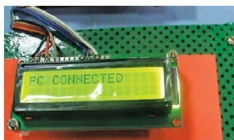


Fig. 10: PC connected

button in Form3. This buzzer gives an indication that the item presented has been disqualified.

The above process is repeated for the next team of the same item by pressing S1 reset button on the micro-controller. However, if item is changed then you need to press reset switch S1 and then master stop switch S2. ●

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